

5. What do you observe?

When n is small (like 500), the bars jump up and down more, and the shape is a bit noisy.

As n gets larger, the histogram becomes smoother and looks more and more like the theoretical triangle (lowest probability at 2 and 12, highest at 7).

So, with big n , the simulated frequencies are very close to the real probabilities.

6. How is this related to “regression to the mean”? (student-style answer)

At the beginning, with a few throws, some sums can appear too often or too rarely (an “extreme” result).

When we keep throwing more dice (increase n), these extreme results are balanced out, and the frequencies move back towards the average / expected values.

This “moving back towards the true probabilities” is exactly what we mean by regression to the mean.